

Carly M. Venenciano

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Data analyst with a B.A. in Mathematics and experience cleaning and validating large multi-file datasets, building end-to-end data infrastructure (PostgreSQL, ETL pipelines, Flask APIs, interactive dashboards), and presenting findings to non-technical audiences. Modeled MLB minor league pitch-tracking and bat-tracking data to identify drivers of hitter success.

EDUCATION

B.A. Mathematics, Minor in Computer Science | Occidental College

May 2025

GPA: 3.71, Cum Laude, Dean's List, COSMOS 2.0 Scholar | Study Abroad: Yonsei University, South Korea

Vice President, Asian Pacific Islander Association (2022-2025)

SKILLS

Languages & Data: SQL (PostgreSQL), Python (Pandas, NumPy), R

Data Engineering: ETL pipelines, Flask (REST APIs), API design and web development (HTML/CSS, JavaScript), shell scripting, Linux, AWS (EC2), Git/GitHub

Analysis & Visualization: Regression/statistical modeling, data cleaning and validation, Vega-Lite, Plotly, Matplotlib, Jupyter

DATA & ANALYTICS EXPERIENCE

Data Analyst | Women in Sports Data Hackathon 2024

Jun 2024 - Jul 2024

- Collected, cleaned, and standardized 1,200+ JSONL files of MLB minor league pitch-tracking and bat-tracking data using Pandas and NumPy
- Designed a custom "success-at-one-pitch" metric and built 8 multilinear regression models (batting, all pitches, six pitch types), using backward selection with VIF and p -value diagnostics to test for multicollinearity and validate feature selection
- Found launch-angle sweet spot % to be the strongest batting predictor, with shorter swing length and higher exit velocity also retained (adj. $R^2 = 0.42$, $n = 31$ batters)
- Documented limitations (small per-pitch-type samples, no at-bat-level data) and recommended next steps in a written report for non-technical readers

Software Engineer Research Intern | Carnegie Mellon University REUSE

May 2023 - Aug 2023

- Designed a PostgreSQL database integrating 2,000+ rows across 18 metrics from multiple WebAssembly engine benchmarks
- Built automated ETL pipelines in Python and shell scripting to load and structure benchmark data
- Developed a Flask API and website with interactive Vega-Lite dashboards that explain the study, define key terms, and present results so non-technical audiences can explore findings self-serve, without running code
- Continued research independently after internship, running WebAssembly benchmarking experiments on AWS EC2
- Presented findings in a poster session and written report

UX/Analytics Project | Soccer Website – Human Computer Interaction Course

Apr 2025

- Created Matplotlib 3D data visualization of top 8 Champions League scorers comparing number of goals with minutes played and touches on the ball; results showed no clear relationship between touches on the ball and goals scored
- Prototyped responsive website in Figma to improve fan engagement and accessibility in soccer analytics

PROFESSIONAL EXPERIENCE

Personal Loan Specialist | OneMain Financial

Sep 2025 - Present

- Ranked 1st among district employees for Q1 and Q2; achieved all performance goals for 3 consecutive months
- Manage sensitive financial and client data with high attention to detail and compliance standards

Math Department Peer Tutor | Occidental College

Aug 2023 - May 2025

- Translated complex mathematical concepts into clear explanations tailored to individual learning styles